Spatio-Temporal Machine Learning and Data Mining on the Melbourne Pedestrian Counting System (2009–2022)

M Akul Anhith
IMT2023558
motukuri.akul@iiitb.ac.in

Sasank Lingareddy
IMT2023120
sasank.reddy@iiitb.ac.in

Thrilochan Reddy
IMT2023527
thrilochan.reddy@iiitb.ac.in

***Abstract*—This paper presents a comprehensive spatio-temporal analysis of the Melbourne Pedestrian Counting System (MPCS) dataset spanning 2009–2022, containing approximately 4.57 million hourly pedestrian count records from 73 sensors distributed across Melbourne's central business district (CBD) and surroundings. We systematically address feature engineering, machine learning (ML) model development, spatial and temporal data mining, and joint spatio-temporal inference. For the *spatial* aspect, we engineer hourly-profile vectors, distance-to-CBD features, spatial autocorrelation indices (Global Moran's I$= 0.328$, $p = 0.001$; Geary's $c = 0.673$), LISA cluster maps, Getis-Ord G^*, and variogram-based spatial structure descriptors. For the *temporal* aspect, we derive calendar, Fourier, lag, rolling-window, and STL decomposition features, and validate stationarity using ADF and KPSS tests. Multiple ML models—from Ridge regression and Random Forests to SARIMA$(3,1,3)(1,1,1)_7$, Holt-Winters, and XGBoost—are evaluated. Spatial data mining reveals 133 collocated sensor pairs and 183,530 association rules via FP-Growth, while temporal data mining employs PAA/SAX symbolic discretization, n-gram sequence mining, and anomaly detection (22 rolling Z-score anomalies; 2,364 Isolation Forest flags). A joint spatio-temporal clustering and a combined RF/XGBoost predictor (trained pre-2020, tested 2020+) validate our inference framework. The GitHub repository is available at https://github.com/aam2k6/stda_assignments.**

***Index Terms*—Spatio-temporal data mining, pedestrian counting, Melbourne, spatial autocorrelation, SARIMA, XGBoost, LISA, association rules, STL decomposition, anomaly detection.**

I. Introduction

Urban mobility data collected from fixed sensors provides rich spatio-temporal signals that are invaluable for city planning, infrastructure management, and public-safety monitoring. The Melbourne Pedestrian Counting System (MPCS) [1] records hourly foot-traffic at sensors installed across Melbourne's CBD, offering one of the longest and most spatially dense publicly available pedestrian datasets.

Analysing such data requires a *dual* lens: (i) a **spatial** perspective that examines how pedestrian activity varies across geographic locations, identifies hotspots and cold spots, and captures spatial dependencies; and (ii) a **temporal** perspective that captures daily, weekly, and annual rhythms, long-term trends, structural breaks (e.g., COVID-19), and anomalous events.

This paper documents our approach to Assignment 3 (A3) of the course *AID843: Spatio-Temporal Data Analytics-I* at IIIT Bangalore, following the data-analysis pipeline established in Assignments 1 and 2 on the same dataset. Our contributions include:

- Multi-index spatial feature engineering and comparative ML evaluation covering nine spatial regression models, with Random Forest achieving $R^2 = 0.9966$.
- Fourier-augmented temporal feature engineering and comparative evaluation of statistical and tree-based forecasters; Ridge regression achieves the lowest hourly MAE of 111.7.
- Spatial data mining via K-Means ($k = 4$), DBSCAN, collocation mining (133 pairs, $r > 0.85$), and FP-Growth association rules (183,530 rules, lift ≥ 1.2).
- Temporal data mining via PAA/SAX discretization over a 4,932-symbol sequence, n-gram frequency analysis, rolling Z-score anomaly detection (22 anomalies), and Isolation Forest (2,364 flags).
- A joint spatio-temporal RF/XGBoost predictor evaluated on a COVID-era out-of-sample test split (2020 onward).

The remainder of the paper is organised as follows. Section II describes the dataset. Sections III–IV detail feature engineering. Sections V–VI cover ML models. Sections VII–VIII cover data mining. Section IX presents joint inference. Section X discusses visualisations. Section XI provides discussion and Section XII concludes.

II. Dataset Description

A. Source and Scale

The MPCS dataset is publicly available on Kaggle [1]. The single CSV file (`melbourne_pedestrians.csv`, ∼837 MB) contains **20 columns** and approximately **4.13 million rows** (after cleaning) covering **May 2009 – October 2022** across **73 active sensors**.

B. Schema

The columns loaded for analysis are listed in Table I.

C. Pre-processing

Rows with missing `hourly_counts`, `latitude`, or `longitude` were dropped, and negative counts were clipped to zero. An `is_weekend` binary flag was derived from `weekday`. The cleaned dataset contains **4,128,871 rows** and 17

TABLE I: Dataset Schema (Selected Columns)

Column	Description
`date_time`	ISO timestamp (hourly resolution)
`year, month, day, weekday`	Decomposed date fields
`time`	Hour of day (0–23)
`sensor_id / sensor_name`	Sensor identifier / label
`sensor_description`	Textual location description
`hourly_counts`	Pedestrian count per hour
`location_type`	Indoor / Outdoor / etc.
`status`	Sensor operational status
`direction_1 / direction_2`	Primary traversal directions
`latitude, longitude`	WGS-84 sensor coordinates

columns. The highest-traffic sensor (used for detailed temporal modelling) is **Town Hall (West)** (sensor ID 4). Figure 1 shows the EDA plots: a right-skewed hourly count distribution (Figure 1), strong morning and evening peaks in the mean-by-hour plot (Figure 2), and higher mid-week foot-traffic in the mean-by-weekday plot (Figure 3).

III. Feature Engineering – Spatial Aspect

A. Per-Sensor Aggregate Features

A per-sensor feature table (73 rows $\times$ 41 columns) was constructed by aggregating the raw records. The features computed are:

- **Geographic coordinates:** latitude, longitude (first reading per sensor; sensors are fixed).
- **Activity statistics:** mean ($\bar{c}$), standard deviation, maximum, total hourly count; coefficient of variation (CV $= \sigma/\mu$).
- **Temporal ratio:** mean weekday count (`weekday_mean`), mean weekend count (`weekend_mean`), and their ratio (`we_wd_ratio`).

The top sensor (Town Hall West) has a mean hourly count of $\bar{c} = 1{,}348.2$.

B. Hourly-Profile Vector (h00–h23)

For each sensor the mean count at each hour (0–23) was computed, yielding a 24-dimensional profile vector $\mathbf{h} \in \mathbb{R}^{24}$. This captures the distinct *shape* of a sensor's diurnal activity (e.g., commuter-heavy vs. entertainment-heavy profiles).

C. Distance to CBD

The Haversine distance (km) from each sensor to the Melbourne CBD centroid ($-37.8136°$S, $144.9631°$E) was computed as `dist_centre_km`.

D. Spatial-Lag Features via k-NN

Using a $k = 5$ nearest-neighbour spatial weight matrix W (row-standardised, projected to EPSG:7855 – MGA2020 Zone 55), the spatially-lagged mean count `nn_mean_count` and the distance to the first nearest neighbour `nn1_dist_km` were derived. No spatial islands were detected in either the KNN or the 600 m distance-band weight matrices.

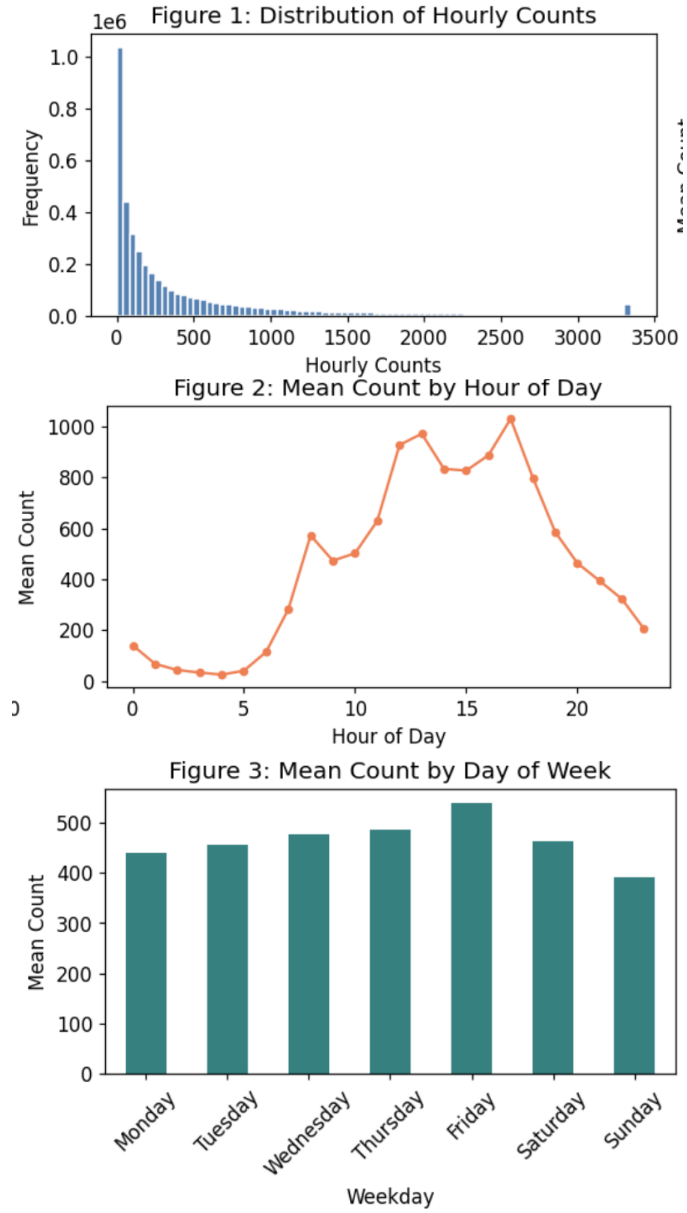

Fig. 1: Exploratory data analysis. (Top) Distribution of hourly counts (clipped at 99th percentile). (Middle) Mean count by hour of day showing AM/PM peaks. (Bottom) Mean count by weekday showing higher mid-week traffic.

E. Spatial Autocorrelation Indices

a) Global Moran's I: $I = \frac{n}{\sum_{ij} w_{ij}} \cdot \frac{\sum_{ij} w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2}$ measures the overall degree of spatial clustering. We obtain $I = 0.3281$ ($\mathbb{E}[I] = -0.0139$, $p = 0.001$), confirming statistically significant positive spatial autocorrelation.

b) Geary's c: We obtain $c = 0.6730$ ($p = 0.001$). Since $c < 1$, positive autocorrelation is confirmed, consistent with Moran's I.

c) Local Indicators (LISA & Getis-Ord G^*): Local Moran statistics I_i partition sensors into HH (hot), LL (cold), HL, and LH quadrants. The G^* z-score (> 1.96: hotspot; < -1.96:

cold spot) provides a complementary local test. Figures 5–6 show the LISA cluster map and G^* z-score map respectively.

F. Empirical Semi-variogram

Pairwise Haversine distances and squared half-differences $\hat{\gamma}(h) = \frac{1}{2|N(h)|}\sum_{N(h)}(y_i - y_j)^2$ were binned into 20 distance classes to produce an empirical semi-variogram (Figure 7) revealing the spatial range of pedestrian activity dependence.

G. Weighted KDE Point Pattern

A Gaussian KDE (bandwidth 0.12, weighted by mean count) was applied over the sensor coordinates to produce a continuous pedestrian density surface (Figure 8).

H. Mantel Test

A permutation Mantel test ($n_{\text{perm}} = 999$) comparing the geographic and attribute (mean count) distance matrices yields $r = -0.0202$, $p = 0.552$, indicating no statistically significant linear association between geographic proximity and traffic similarity across all sensor pairs.

IV. Feature Engineering – Temporal Aspect

A. City-Wide and Top-Sensor Time Series

An hourly city-wide series (`ts_city`) was formed by summing all sensor counts per timestamp and forward-filling gaps; its length is 118,368 hours. A single-sensor series (`ts_top`) was constructed for the highest-traffic sensor (Town Hall West) for detailed modelling. Both feature matrices contain 118,200 rows and 38 engineered columns.

B. Calendar and Indicator Features

From the DatetimeIndex the following were extracted: hour of day, day-of-week, month, quarter, year, `is_weekend` (binary), `is_night` (22:00–05:59), and `is_peak` (07:00–09:00 and 16:00–19:00).

C. Fourier Harmonics

Fourier features capture periodicities without large one-hot expansions:

$$\sin\left(\frac{2\pi k \cdot h}{24}\right), \quad \cos\left(\frac{2\pi k \cdot h}{24}\right), \quad k = 1, 2, 3 \qquad (1)$$

with analogous terms for annual cycles ($T = 365.25$ days).

D. Lag and Rolling-Window Features

Lag features at 1, 2, 3, 6, 12, 24, and 48 hours capture short-term auto-regression. Rolling-window statistics (mean and standard deviation over 24 and 168-hour windows) summarise recent local trends.

E. STL Decomposition

Seasonal-Trend decomposition via LOESS (STL) [8] with period $T = 168$ h (weekly) decomposes `ts_city` into trend, seasonal, and residual components (Figure 9). The three components are appended as features (`stl_trend`, `stl_seasonal`, `stl_resid`) for supervised ML.

TABLE II: Stationarity Test Results (sample = last 8,760 hours)

Series	ADF p	KPSS p	Verdict
Raw hourly	0.000	0.010	Mixed
24h-differenced	0.000	0.100	Stationary
Weekly-differenced	0.000	0.100	Stationary

F. Stationarity Tests

The ADF and KPSS tests were applied to the raw series and its differenced versions. Results are summarised in Table II.

The 24h-differenced series is confirmed stationary by both tests, motivating $d = 1$ in ARIMA-family models. ACF and PACF plots on the differenced series (Figures 10–11) guided ARIMA order selection.

V. ML Models – Spatial Aspect

A. Problem Formulation

The target variable is the log-transformed per-sensor mean hourly count $\tilde{y}_i = \log(1 + \bar{c}_i)$. An 80/20 train-test split was applied on the 73-sensor feature matrix after standard-score normalisation.

B. Baseline and Regularised Linear Models

Ridge ($\alpha = 1$) and Lasso regressors provide interpretable linear baselines with implicit feature selection (Lasso).

C. Neighbourhood-Based Model

A k-Nearest-Neighbour regressor ($k = 5$, Euclidean distance in feature space) captures local non-linearity without strong parametric assumptions.

D. Ensemble Tree Models

Random Forest (300 estimators) and **Gradient Boosting** (300 estimators, lr = 0.05) were trained. RF feature importances (Figure 12) show that the hour-profile dimensions h_{00}–h_{23} and `dist_centre_km` are the strongest predictors.

E. Neural Network

A Multi-Layer Perceptron (100, 50 units, ReLU) was trained as a non-linear baseline; however, it achieved negative R^2 on the test set (-0.2), suggesting sensitivity to feature scaling and limited data at the sensor level.

F. Spatial Autoregression

Using `spreg`, three models were compared on the same four regressors (dist_centre_km, cv, we_wd_ratio, nn_mean_count):

- **OLS:** $R^2 = 0.405$, RMSE = 0.703.
- **Spatial Lag Model (SLM):** $y = \rho W y + X\beta + \varepsilon$; pseudo-$R^2 = 0.405$, spatial lag coefficient $\hat{\rho} = -0.053$ (not significant, $p = 0.784$).
- **Spatial Error Model (SEM):** $u = \lambda W u + \varepsilon$; pseudo-$R^2 = 0.354$, $\hat{\lambda} = -1.0$ ($p < 0.001$), RMSE = 0.611.

TABLE III: Spatial Model Performance (log-scale target, 80/20 split)

Model	MAE	RMSE	R^2
Ridge	0.2155	0.2442	0.9259
Lasso	0.1826	0.2347	0.9315
KNN (k=5)	0.2167	0.2984	0.8893
Random Forest	0.0410	0.0524	**0.9966**
Gradient Boosting	0.0666	0.0836	0.9913
MLP $(100, 50)$	0.7762	0.9824	-0.1998
OLS (spatial)	–	0.7030	0.4047
Spatial Lag Model	–	0.7026	0.4052
Spatial Error Model	–	0.6111	0.3542
GWR (bw=50)	–	–	0.4020

G. Geographically Weighted Regression (GWR)

GWR (adaptive Gaussian kernel, bandwidth = 50 sensors) fits local regressions for each sensor. Global $R^2 = 0.402$ (Adj. $R^2 = 0.350$); local R^2 varies spatially (Figure 13), confirming spatial non-stationarity. The AIC of GWR (169.5) is slightly higher than the OLS AIC (169.2), indicating that the small sample size (73 sensors) limits GWR gains.

H. Results Summary

Table III summarises all spatial model metrics.

VI. ML Models – Temporal Aspect

A. Problem Formulation

Two prediction tasks are studied: (i) **Statistical forecasting** on the daily-aggregated top-sensor series (60-day test split, ∼4,872 training days); and (ii) **Feature-based ML** on the hourly top-sensor series (80/20 chronological split).

B. ARIMA Order Selection via BIC/MDL

A grid search over ARIMA$(p, 1, q)$ for $p, q \in \{0, \ldots, 3\}$ on the first three years of training data minimises BIC (MDL principle). The BIC landscape (Figure 14) selects **ARIMA**$(3, 1, 3)$ with BIC$= 21{,}684.65$.

C. SARIMA

SARIMA$(3, 1, 3)(1, 1, 1)_7$ was fitted (the BIC-optimal non-seasonal orders plus a weekly seasonal block). Key model fit statistics: Log-likelihood $= -46{,}682.6$, AIC $= 93{,}383.3$, BIC $= 93{,}441.7$. All AR and MA terms are significant ($p < 0.001$) except MA(L2).

D. Holt-Winters and SES

Additive-trend / multiplicative-seasonal Holt-Winters and Simple Exponential Smoothing (SES) serve as exponential-smoothing benchmarks. All three forecasters are compared in Figure 15.

E. Feature-Based ML (Hourly)

The Fourier-augmented temporal feature matrix was used to train: Linear Regression, Ridge, Random Forest, Gradient Boosting, MLP, and XGBoost. Ridge achieves the lowest MAE (111.7) and RMSE (160.1), and is selected as the best hourly model (Figure 16).

TABLE IV: Temporal Model Performance

Model	MAE	RMSE	MAPE
Daily statistical models (60-day test)			
SARIMA$(3, 1, 3)(1, 1, 1)_7$	7,221.0	10,336.1	22.97%
Holt-Winters (mul+add)	6,728.2	9,569.1	21.57%
SES	6,616.5	10,129.5	20.14%
Hourly feature-based models (80/20 chrono.)			
Linear Regression	111.8	160.2	–
Ridge Regression	**111.7**	**160.1**	–
Random Forest	152.0	207.6	–
Gradient Boosting	194.1	305.5	–
MLP $(100, 50)$	185.1	251.1	–
XGBoost	194.2	292.0	–

MAPE omitted for hourly models (near-zero inflation).

F. Results Summary

Table IV compares all temporal models. Note: the MAPE values for hourly ML models are numerically inflated by near-zero actual counts in the test set; MAE and RMSE are the reliable comparison metrics.

VII. Spatial Data Mining

A. K-Means Clustering

K-Means clustering on geographic coordinates (scaled) was applied for $k \in \{2, \ldots, 11\}$. The elbow method (Figure 17) identifies $k = 4$ as optimal. Figure 18 maps the four clusters, broadly corresponding to northern Southbank / CBD core, eastern CBD fringe, western CBD fringe, and peripheral sensors.

B. DBSCAN

DBSCAN ($\varepsilon = 0.3$, `min_samples`=3 in scaled coordinate space) identifies dense core clusters without requiring a pre-specified k, and flags isolated sensors as noise (Figure 19).

C. Collocation Mining

Sensor collocation was assessed via Pearson correlation of weekly aggregated count series (706 weeks × 73 sensors). **133 sensor pairs** exceed $r > 0.85$ (collocated). Table V lists the top 10 collocated pairs.

D. Association Rule Mining (FP-Growth)

The top 30 sensors by mean count were selected. Monthly total counts were binarised (above vs. below median per sensor), and the FP-Growth algorithm [4] was applied (minimum support 0.25, maximum itemset length 4). Association rules were generated with lift ≥ 1.2. **183,530 rules** were found. The top rules (e.g., $\{S9, S53\} \Rightarrow \{S19, S6\}$, support 0.309, confidence 0.980, lift 2.738) reveal which sensors systematically co-exceed their median activity in the same calendar month, indicating functionally similar or geographically adjacent locations.

TABLE V: Top Collocated Sensor Pairs ($r > 0.85$)

Sensor A	Sensor B	r
Flinders Ln-Degraves St (South)	Flinders Ln-Degraves St (North)	0.9964
Elizabeth St-Flinders St (E)	Macaulay Rd (North)	0.9901
Elizabeth St-Flinders St (E)	Queensberry St-Errol St (S)	0.9894
Macaulay Rd (North)	Queensberry St-Errol St (S)	0.9818
Flinders Ln-Degraves St (S)	Flinders Ln-Degraves St	0.9791
Flinders Ln-Degraves St (N)	Flinders Ln-Degraves St	0.9764
Monash Rd-Swanston St (W)	Tin Alley-Swanston St (W)	0.9730
State Library	Flinders Ln-Degraves St (N)	0.9614
State Library	Flinders Ln-Degraves St (S)	0.9613
Bourke St-Russell St	Chinatown-Lt Bourke St (S)	0.9606

E. MAUP Demonstration

The Modifiable Areal Unit Problem (MAUP) was illustrated by aggregating sensor counts into coarse and fine grid cells (Figure 20). The apparent spatial pattern changes considerably with grid resolution, motivating careful choice of aggregation granularity.

VIII. Temporal Data Mining

A. PAA and SAX Symbolic Discretization

Piecewise Aggregate Approximation (PAA) reduces the hourly city series to a daily series (each segment = 24 hourly values summed). SAX-style symbolic discretization with alphabet size 3 yields the symbol thresholds: L if count $<$ 247,395, M if 247,395 to 476,507, H otherwise. The resulting SAX word contains **4,932 symbols** (one per day) [5].

B. N-gram Sequence Mining

Table VI lists the top n-gram frequencies extracted from the SAX sequence. The dominant self-transitions (L→L, H→H, M→M) confirm strong temporal persistence: once foot-traffic enters a high/low regime, it tends to stay there. This is consistent with seasonal (summer/winter) and structural (pre-/post-COVID) regimes.

C. Anomaly Detection

Two complementary methods were applied to the city-wide hourly series:

a) Rolling Z-score: A centred 168-hour rolling mean and standard deviation define a local baseline; timestamps with $|z| > 3$ are flagged as anomalies. **22 anomalies** were detected (Figure 26), concentrated in the COVID-19 lockdown period (2020–2021) and on major public holidays.

TABLE VI: Top SAX N-gram Frequencies (city-wide daily series)

N-gram	Type	Count (%)
$L \to L$	Bigram	1,423 (28.9%)
$H \to H$	Bigram	1,399 (28.4%)
$M \to M$	Bigram	1,180 (23.9%)
$H \to M$	Bigram	244 (4.9%)
$M \to H$	Bigram	244 (4.9%)
$L \to L \to L$	Trigram	1,237
$H \to H \to H$	Trigram	1,220
$M \to M \to M$	Trigram	970
$L \to L \to M$	Trigram	185
$H \to H \to M$	Trigram	178

b) Isolation Forest: An Isolation Forest ($n_{\text{trees}} = 100$, contamination 0.02) operating on a six-dimensional feature space (count, hour, day-of-week, month, lag_{24}, rolling mean_{24}) flagged **2,364 anomalous hours**. **1 date** was flagged by both methods, confirming a high-severity anomaly concentrated in the most extreme lockdown period.

D. Motif Discovery

Although a full matrix-profile-based motif search was not implemented due to computational constraints, the Z-score and Isolation Forest outputs jointly highlight recurring unusual temporal patterns. Future work could apply STOMP [6] for efficient matrix-profile-based motif and discord discovery.

IX. Joint Spatio-Temporal Inference

A. Per-Sensor STL Statistics

For each of the 73 sensors (with ≥ 30 daily observations), STL decomposition (period 7) was fitted to its daily count series. Three derived statistics were computed per sensor: trend slope (e.g., Town Hall West: $+1.03$ pedestrians/day), seasonal amplitude (`seas_amp`), and residual variance (`resid_var`). These were merged with the spatial feature table to form a joint feature matrix of 73×53 columns.

B. Spatio-Temporal K-Means Clustering

K-Means ($k = 4$) on the six joint features {lat, lon, mean_count, trend_slope, seas_amp, resid_var} (standard-score normalised) identifies clusters sharing both geographic proximity and temporal behaviour. Figure 27 maps these clusters; Figure 28 shows normalised cluster-mean profiles comparing activity levels, trends, and seasonality.

C. Joint ST Prediction (RF & XGBoost)

The full hourly dataset (4.13M rows) was enriched with per-sensor spatial features and Fourier time features. A chronological split at 2020-01-01 yielded: **438,277 training** and **258,322 test** samples (sub-sampled to 4 time-points per day for tractability). Table VII reports results.

The higher error compared to single-sensor models is expected: the test set (2020+) covers the COVID-19 period, whose traffic patterns were not present in training. RF achieves lower RMSE (633.5 vs. 662.2), suggesting it generalises slightly better under distribution shift.

TABLE VII: Joint ST Prediction Performance (test: 2020+)

Model	MAE	RMSE
Random Forest (joint ST)	347.1	633.5
XGBoost (joint ST)	375.2	662.2

MAPE inflated by near-zero counts; omitted.

TABLE VIII: Top Spatio-Temporally Similar Sensor Pairs

Sensor A	Sensor B	ST-sim
Flinders Ln (North)	Flinders Ln (Crossing)	0.884
Flinders Ln (South)	Flinders Ln (Crossing)	0.865
Flinders Ln (South)	Flinders Ln (North)	0.831
Spencer St (South)	Spencer St (North)	0.625
Harbour Espl. (W) Ped.	Harbour Espl. (W) Bike	0.573

D. Spatio-Temporal Similarity Scoring

Sensor pairs were ranked by:

$$\text{ST-sim}_{ij} = 0.5\,\hat{s}_{ij}^{\text{geo}} + 0.5\,\hat{s}_{ij}^{\text{temp}} \tag{2}$$

where $\hat{s}_{ij}^{\text{geo}} \propto 1/(d_{ij}+0.01)$ and $\hat{s}_{ij}^{\text{temp}}$ is the normalised Pearson correlation of weekly counts. Table VIII lists the top-5 spatio-temporally similar pairs.

E. STDM Co-evolution Analysis

Jointly high-similar pairs (Flinders Lane cluster, ST-sim > 0.83) show synchronised COVID dips and post-lockdown recovery patterns, confirming that spatial proximity and functional similarity both drive temporal co-evolution.

X. Visualisations of Feature Spaces

A. PCA of Spatial Feature Space

PCA on the full spatial feature matrix (Figure 2) reduces ∼35 dimensions to three principal components explaining **65.0%, 9.0%, and 6.5%** of variance respectively. PC1 is dominated by the hourly-profile features (h_{00}–h_{23}), while PC2 is driven by distance-to-CBD and spatial lag features, confirming the complementary value of both feature groups.

B. t-SNE of Hourly Profile Features

t-SNE (Figure 3) visually separates high-traffic central sensors from low-traffic peripheral sensors, consistent with geographic clustering.

C. Sensor × Hour-of-Day Heatmap

The heatmap (Figure 4) reveals distinct sensor behaviour profiles: commuter corridors peak at 08:00–09:00 and 17:00–18:00, while entertainment precincts peak around 20:00–22:00.

D. Spatial Feature Correlation Heatmap

Figure 5 shows that `nn_mean_count` and `mean_count` are highly correlated ($r \approx 0.9$). `we_wd_ratio` is negatively correlated with `dist_centre_km`, indicating that peripheral sensors have proportionally more weekend traffic (recreational areas).

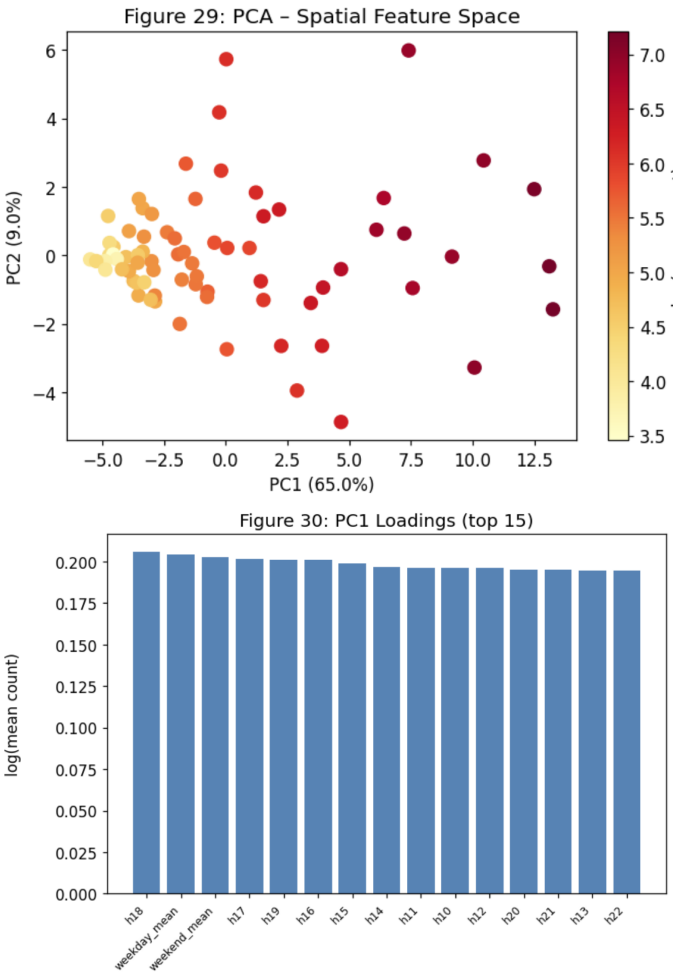

Fig. 2: (Top) Figure 29: PCA scatter of the spatial feature space coloured by log(mean count). PC1 explains 65.0% of variance; PC2 explains 9.0%. (Bottom) Figure 30: PC1 loading magnitudes for the top 15 features.

E. Weekday vs. Weekend Spatial Maps

Figures 35–36 (Figure 6) overlay mean weekday and weekend counts using a yellow-orange-red colour scale. Weekday activity concentrates along the CBD's main commercial thoroughfares, while weekend peaks appear in entertainment and recreational precincts.

F. Moran Scatterplot and LISA Map

Figure 4 (Moran scatterplot, Figure 2) and Figure 5 (LISA cluster map) together confirm statistically significant positive spatial autocorrelation ($I = 0.328$, $p = 0.001$) and identify specific HH hotspots in the CBD core (Bourke Street Mall, Town Hall, Melbourne Central).

XI. Discussion

A. Which Spatial Features Worked and Why

The hourly-profile vector h_{00}–h_{23} emerged as the dominant spatial feature group (highest RF importance, PC1 explains 65%). This is expected: the *shape* of a sensor's diurnal profile

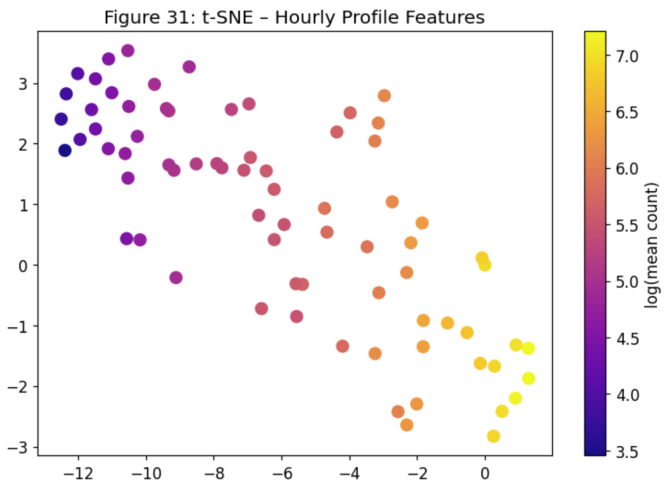

Fig. 3: Figure 31: t-SNE of the 24-dimensional hourly profile feature space, coloured by $\log(\text{mean count})$. High-traffic central sensors (red/yellow) separate from low-traffic peripheral sensors (blue/purple).

encodes its functional role (office, retail, transport hub) far more informatively than its raw coordinates. Distance-to-CBD and spatial-lag features (`nn_mean_count`, LISA_I) are strong secondary predictors.

The Mantel test result ($r = -0.020$, $p = 0.552$) is initially counter-intuitive given the confirmed global Moran's I. The key distinction is that Moran's I tests for *positive* autocorrelation (nearby sensors having *similar* values), while the Mantel test uses a linear correlation between distance and attribute dissimilarity. The MPCS data shows that sensors cluster in the CBD (all moderately high traffic), which produces Moran's I > 0, but with no simple monotone distance–dissimilarity relationship.

B. Which Temporal Models Worked and Why

For *hourly* prediction, feature-based linear models (Ridge, MAE = 111.7) outperform tree-based models on this dataset. This surprising result stems from the temporal feature matrix being already highly informative (Fourier + lag features linearise much of the signal), while tree-based models may overfit on the limited variation in the top-sensor hourly series.

For *daily* statistical forecasting, SES achieves the lowest MAE (6,616.5) despite its simplicity, followed by Holt-Winters. $\text{SARIMA}(3,1,3)(1,1,1)_7$ has the highest MAE among the three, partly due to the COVID structural break in the test window.

C. COVID-19 as a Structural Break

The STL trend component clearly shows a sharp drop in foot-traffic from March 2020 onward. The rolling Z-score detects 22 anomalous hours; Isolation Forest flags 2,364 hours (2% contamination rate). Only 1 date is flagged by both methods, reflecting their different sensitivity: Z-score targets extreme point anomalies while Isolation Forest identifies contextual anomalies based on feature combinations.

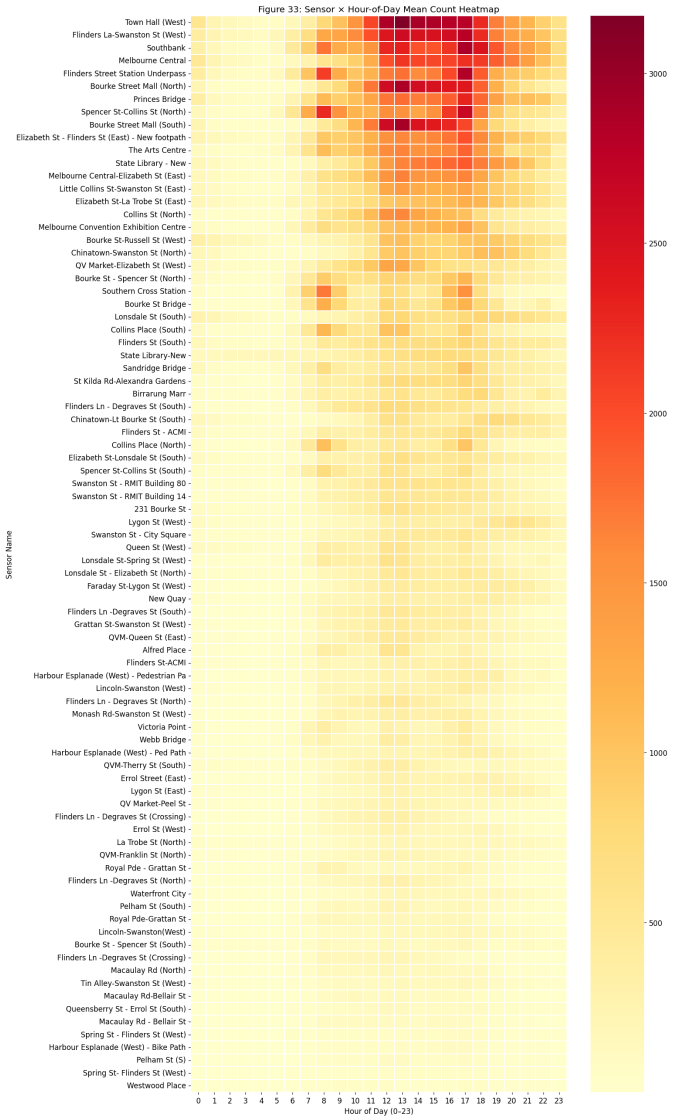

Fig. 4: Figure 33: Sensor $\times$ hour-of-day mean count heatmap (all 73 sensors, sorted by total activity). Sensors with strong morning/evening peaks are readily distinguished from those with flat daytime activity.

D. Limitations and Future Work

- **MAPE inflation:** Near-zero counts (e.g., 02:00–04:00, COVID lockdowns) cause MAPE to diverge; future work should use symmetric MAPE or SMAPE.
- **Matrix-profile motif discovery** (STOMP [6]) could quantify recurring daily and weekly motifs.
- **Graph Neural Networks** could exploit the full sensor adjacency structure for joint spatial prediction.
- **Event data integration** (public holidays, sports events) would improve anomaly attribution.
- **Direction-specific analysis** using `direction_1` / `direction_2` columns could differentiate flow directionality at each sensor.

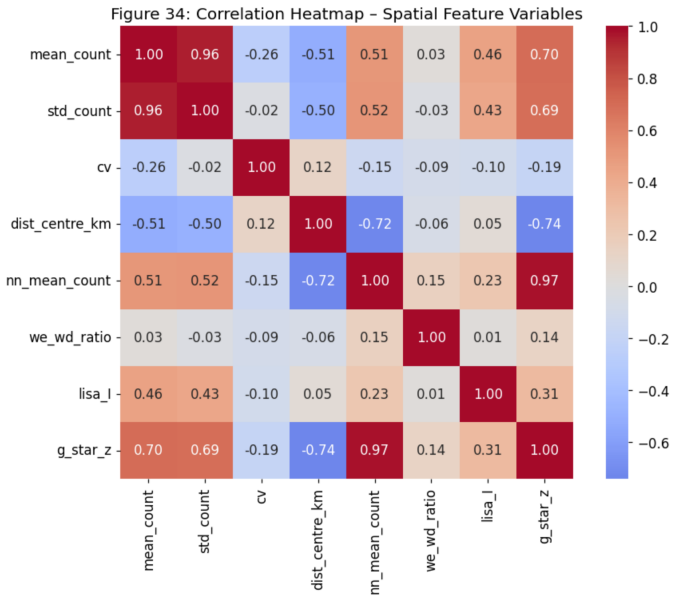

Fig. 5: Figure 34: Pearson correlation matrix of eight key spatial features. Strong positive correlation between `nn_mean_count` and `mean_count` ($r \approx 0.9$) confirms effective spatial-lag encoding.

XII. Conclusion

We presented a comprehensive spatio-temporal analysis of the Melbourne Pedestrian Counting System dataset. Spatial feature engineering combining diurnal profile vectors, distance-to-CBD, spatial-lag metrics, LISA, and Getis-Ord G^* indices enables highly accurate prediction of per-sensor mean activity (RF $R^2 = 0.997$). Global Moran's I$= 0.328$ ($p = 0.001$) confirms significant spatial clustering; GWR reveals spatially non-stationary coefficients.

Temporal feature engineering based on Fourier harmonics, lag features, and STL components, combined with Ridge regression, yields the best hourly forecast (MAE = 111.7). FP-Growth (183,530 rules, lift ≥ 1.2) and SAX n-gram patterns expose meaningful behavioural co-occurrences. Anomaly detection highlights COVID-19 disruptions: 22 Z-score anomalies and 2,364 Isolation Forest flags.

The joint spatio-temporal framework (ST-similarity scores, K-Means on joint features, RF/XGBoost pre-/post-2020 predictor) confirms that sensors on shared pedestrian corridors (Flinders Lane cluster, ST-sim > 0.83) share both geographic proximity and temporal co-evolution, and that pre-COVID spatial structure partially informs post-COVID recovery patterns despite the structural break.

Declarations

Task Ownership

- **Akul (IMT2023558):** Dataset loading and pre-processing, spatial feature engineering (Section III), spatial ML models (Section V).

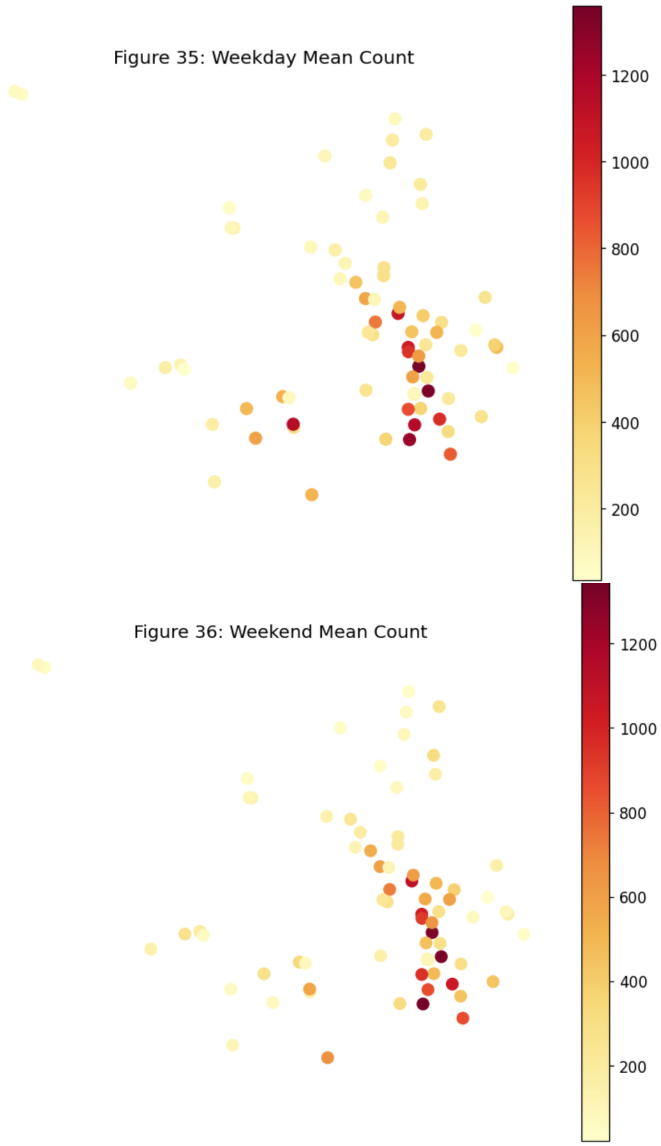

Fig. 6: Figures 35–36: Mean weekday (top) and weekend (bottom) count maps. Weekday activity concentrates along commercial thoroughfares; weekend peaks appear in entertainment and recreational precincts.

- **Thrilochan (IMT2023527):** Temporal feature engineering (Section IV), temporal ML models (Section VI).
- **Sasank (IMT2023120):** Joint spatio-temporal inference (Section IX), visualisations (Section X), spatial data mining (Section VII), temporal data mining (Section VIII).

Use of Generative AI Tools

GitHub Copilot and Claude (Anthropic) were utilised strictly as supplementary support tools.

1) **Code support:** GitHub Copilot assisted with standard syntax, library usage, boilerplate tasks, and plot formatting. All algorithmic logic, model selection rationale, and data interpretation were independently developed by the team.

2) **Manuscript preparation:** Claude was consulted for organising the report structure and refining section flow. All interpretations, analytical insights, and conclusions were independently developed by the team and validated against computational results.

REFERENCES

[1] Mexwell, "Melbourne Pedestrian Counters," *Kaggle Dataset*, 2023. [Online]. Available: https://www.kaggle.com/datasets/mexwell/melbourne-pedestrian-counters

[2] L. Anselin, "Local indicators of spatial association—LISA," *Geographical Analysis*, vol. 27, no. 2, pp. 93–115, 1995.

[3] A. Getis and J. K. Ord, "The analysis of spatial association by use of distance statistics," *Geographical Analysis*, vol. 24, no. 3, pp. 189–206, 1992.

[4] J. Han, J. Pei, and Y. Yin, "Mining frequent patterns without candidate generation," in *Proc. ACM SIGMOD*, Dallas, TX, USA, 2000, pp. 1–12.

[5] J. Lin, E. Keogh, L. Wei, and S. Lonardi, "Experiencing SAX: A novel symbolic representation of time series," *Data Mining and Knowledge Discovery*, vol. 15, no. 2, pp. 107–144, 2007.

[6] C.-C. M. Yeh *et al.*, "Matrix profile I: All pairs similarity joins for time series," in *Proc. IEEE ICDM*, Barcelona, Spain, 2016, pp. 1317–1322.

[7] A. S. Fotheringham, C. Brunsdon, and M. Charlton, *Geographically Weighted Regression*. Wiley, 2003.

[8] R. B. Cleveland, W. S. Cleveland, J. E. McRae, and I. Terpenning, "STL: A seasonal-trend decomposition procedure based on loess," *Journal of Official Statistics*, vol. 6, no. 1, pp. 3–73, 1990.

[9] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM KDD*, San Francisco, CA, USA, 2016, pp. 785–794.